Paleomagnetic study of lower Mesozoic diabase dikes and sills of Connecticut and Maryland

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Paleomagnetic results from 61 sites on diabase dikes and sills in Connecticut and Maryland yield a mean pole of position of 100.9° E, 68.6° N, $\alpha_{95} = 1.6°$. These results combined with those from 78 sites on four diabase sills in the Gettysburg Basin of Pennsylvania produce a mean pole position of 101.6° E, 65.4° N, $\alpha_{95} = 1.3°$ from 139 VGPs. These rocks are probably of Early to early Middle Jurassic age by stratigraphic and inferred stratigraphic position. The paleomagnetic results indicate that the Gettysburg and Hartford Basins did not subside simultaneously.

The dispersion of virtual geomagnetic poles about the mean pole of the Connecticut, Pennsylvania, and Maryland intrusive rocks is probably the result of 'homogenization' of magnetic directions during low-temperature oxidation of titanomagnetite to titanomaghemite over a period long enough to erase a considerable amount of dispersion due to secular variation.

Rotation of Europe against North America in the prerifting configuration causes good convergence of European Jurassic mean poles and the pole of the Connecticut, Maryland, and Pennsylvania intrusive rocks. This is evidence that separation of the two continents had not begun by Early to early Middle Jurassic time, which is in agreement with earlier findings based on sea-floor magnetic anomalies and faunal assemblages.

Les résultats d'études paléomagnétiques faites au Connecticut et au Maryland sur 61 sites de dykes et de filons-couches de diabase donnent une position du pôle moyen de 100.9° E, 68.6° N, $\alpha_{95} = 1.6°$. Ces résultats combinés à ceux pour 78 sites répartis dans quatre filons-couches de diabase du Bassin de Gettysburg en Pennsylvanie donnent une position du pôle moyen de 101.6° E, 65.4° N, $\alpha_{95} = 1.3°$ à partir de 139 VGPs. De par leur position stratigraphique, ces roches datent probablement du Jurassique Inférieur au Jurassique Moyen inférieur. Les résultats paléomagnétiques indiquent que les Bassins de Gettysburg et de Hartford ne se sont pas enfoncés simultanément.

La dispersion des VGPs autour du pôle moyen des intrusifs du Connecticut, de la Pennsylvanie, et du Maryland est probablement le résultat d'une 'homogénéisation' des directions magnétiques durant l'oxydation à basse température de la titanomagnétite en titano-magnémite s'étendant sur une période de temps assez longue pour effacer une quantité considérable de la dispersion causée par la variation séculaire.

La rotation de l'Europe par rapport à l'Amérique du Nord selon la configuration pré-dérive produit une bonne convergence des pôles moyens du Jurassique européen et du pôle des intrusifs du Connecticut, du Maryland, et de la Pennsylvanie. Ceci montre qu'au Jusassique Inférieur à Moyen inférieur la séparation des deux continents n'avait pas commencé, ce qui est en accord avec les conclusions basées sur des anomalies magnétiques du fond océanique et sur des assemblages de faune.

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Introduction

Paleomagnetic results from 61 of 72 sampling sites of Upper Triassic – Lower Jurassic diabase intrusions in Connecticut and Maryland produce a paleomagnetic pole position located at 100.9°E, 68.6°N, $\alpha_{95} = 1.6°$. The sites are located on the Buttress, Higganum, Fairhaven, Cross Rock, and Bridgeport dikes of Connecticut (Fig. 1) and the Boyds sill, southern portion of the Gettysburg sill, and a prominent N–S dike swarm (herein called the Frederick dikes) of central Maryland (Fig. 2). These intrusive rocks were studied for the following reasons:

(1) To determine by paleomagnetic pole positions the age of intrusion of these units relative to the age of tilting of the rocks of the Hartford Basin of Connecticut and the Gettysburg Basin of Pennsylvania and Maryland. DeBoer (1968) established a chronology of dike emplacement for the Hartford Basin. However, since his work was published, sufficient information has appeared concerning paleomagnetism of similar

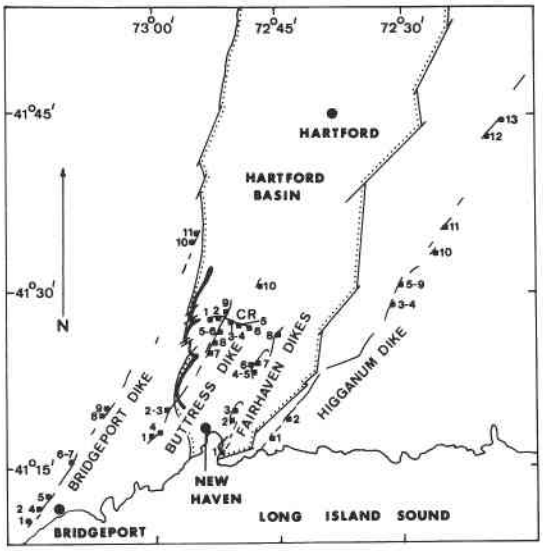


FIG. 1. Site location map of Connecticut. The borders of the Hartford Basin are stippled. Sampling sites are numbered.

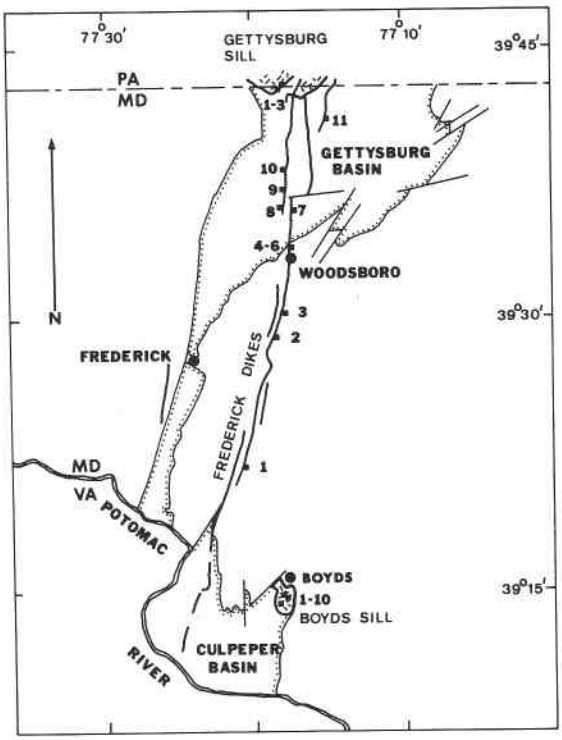


FIG. 2. Site location map of Maryland. The borders of the Gettysburg and Culpeper Basins are stippled. Sampling sites are numbered.

diabase bodies in the Appalachians (Beck 1972), palynology of the Newark Group rocks (Cornet and Traverse 1975), paleosecular variation in the Upper Triassic – Lower Jurassic (Beck 1972), and reconstruction of the Atlantic continents in their pre-rifting fit (Phillips and Forsyth 1972) to make a re-examination of the paleomagnetism of the Connecticut dikes desirable. The only previous paleomagnetic study of the Maryland diabase intrusions was a reconnaisance survey by Bowker (1960).

(2) To produce paleomagnetic data in sufficient quantity to yield information on paleosecular variation (PSV) during the Upper Triassic – Lower Jurassic interval and determine if the observed dispersions of site virtual geomagnetic poles (VGP's) are due to actual PSV or a mineralogic process affecting the magnetic directions of the diabase.

The geology and structural history of the Hartford and Gettysburg Basins have been reviewed by Sanders (1963) and Faill (1973). Sanders concluded that two phases of intrusive activity, separated by tectonic rotation of the basin rocks, occurred in the Hartford Basin. This interpretation is of interest in that paleomagnetic pole positions of rotated and unrotated intrusive bodies should be readily distinguishable. Faill (1973) stated that four large diabase bodies in the Gettysburg Basin (Gettysburg, York Haven, Birdsboro, and Quakertown sills) were intruded before the basin rocks were tilted, as indicated by similar deformation and folding of the sills and their host rocks. Faill's (1973) conclusion was supported by Beck's (1972) paleomagnetic data on these bodies.

Paleomagnetic studies of the Appalachian diabase bodies have been performed by Bowker (1960), Opdyke (1961), Larochelle and Wanless (1966), deBoer (1967, 1968), and Beck (1972) (Table 1). Bowker's (1960) results are of limited value because of incomplete sample cleaning and deBoer's (1967) results are in significant disagreement with those of the other paleomagnetic studies concerning these rocks. The results of the remaining studies are generally in excellent agreement and indicate that the Upper Triassic – Lower Jurassic paleomagnetic pole for North America was located at 87–105°E, 60–70°N. DeBoer (1967) produced pole positions of 125–156°E, 62–73°N from dikes along the length of the Appalachian trend, and subsequently (deBoer 1968) concluded that dikes in central Connecticut

TABLE 1. Paleomagnetic pole positions from Upper Triassic – Lower Jurassic diabase intrusions of the central and northern Appalachians

Location	Sites/Specimens	α_{95}	Long. E	Lat. N	Reference
Pennsylvania					
Quakertown sill*	9/50	4.3	98.5	66.0	Beck (1972)
Birdsboro sill*	26/168	3.2	105.0	62.5	Beck (1972)
York Haven sill*	37/202	1.5	105.5	61.0	Beck (1972)
Gettysburg sill*	13/57	8.3	103.0	65.0	Beck (1972)
New Jersey					
Diabase sills*	5/69	8.6	112.9	64.2	Opdyke (1961)**
Connecticut					
Cheshire dikes†	?/41	4.0	103.0	76.0	deBoer (1968)
West Rock sill*	3/18	4.0	89.0	63.0	deBoer (1968)
Mt Carmel sill*	5/33	4.0	87.0	61.0	de Boer (1968)
Nova Scotia					
Shelburne dike	2/11	4.5	98.0	69.0	Larochelle and Wanless (1966)

NOTE: α_{95} = Radius of cone of 95% confidence. Long. = longitude of paleomagnetic pole. Lat. = latitude of paleomagnetic pole.
*Results from these rock units have been corrected for structure.
**Paleomagnetic pole position calculated from data given in the original reference.
†These units are called the Buttress and Cross Rock dikes in this paper.

could be assigned to Triassic or Jurassic intrusive events on the basis of different paleomagnetic vector directions. Beck (1972) produced a pole position of 105°E, 65°N from 78 sites on the four Gettysburg Basin sills and noted that the dispersion of site VGP's could be explained by PSV of abnormally low magnitude.

Geologic Setting of the Diabase Intrusive Rocks

The Buttress, Higganum, Fairhaven, Bridgeport, and Cross Rock dikes outcrop in central Connecticut, and trend N10–20°E, except for the Cross Rock dike which trends roughly E–W. The Bridgeport and Higganum dikes outcrop entirely in the piedmont west and east of the Hartford Basin, respectively, whereas the Cross Rock and Fairhaven dikes outcrop exclusively in the basin. The basin rocks dip 15–30°E. The Buttress dike outcrops in both the piedmont and basin rocks and transects the peidmont–basin boundary 3 km northwest of New Haven. Both the Buttress and Fairhaven dikes were emplaced in faults (Fritts 1963; deBoer 1968) with trends similar to those cutting lava flows intercalated with the Meriden Formation of the basin, suggesting that no relationship exists between the flows and the Buttress and Fairhaven dikes. The Cross Rock dike was considered by Fritts (1963) to be an extension of the West Rock sill and is crosscut by the Buttress dike, as is the West Rock sill, but aeromagnetic data (Anonymous 1966) suggests that there may be no relationship between the Cross Rock dike and the West Rock sill.

The Gettysburg sill and the Frederick dikes outcrop in the southern part of the Gettysburg Basin of Pennsylvania and Maryland. They are apparently members of the Rossville diabase group (Smith *et al.* 1975), but the Frederick dikes are somewhat younger, as they transect the Gettysburg sill and have chilled margins in the sill. The Gettysburg sill outcrop is restricted to the basin, whose rocks dip 15 to 30° NW. The Frederick dikes outcrop in both the basin rocks and in the piedmont, crossing the boundary between the two near Woodsboro, Maryland. The southern part of the Frederick dikes outcrops in the northern extension of the Culpeper Basin of Virginia and Maryland along with the Boyds sill, a small (about 10 km^2) stock-like body emplaced on the unconformity on the east margin of the Culpeper Basin near Boyds, Maryland. Cloos *et al.* (1964) showed that it is at least 80 m thick and aeromagnetic data (Blanchett *et al.* 1963) indicate that it probably has discordant contacts.

Analytical Techniques

From 5 to 22 25-mm diameter cores were drilled from 72 sites, yielding a total of 448 cores, from which 713 25 × 23 mm specimens were cut. Sampling at each site covered 5–10 m, and wherever possible only fresh artificial exposures were sampled. Specimens were measured

on a Schonstedt SSM-1A spinner magnetometer and demagnetized in a Schonstedt GSD-1 AF demagnetizer. Forty-nine polished sections were prepared and examined with a Reichert Metapan ore microscope at 900 power using oil immersion techniques. A pilot suite of 5–8 specimens from each site was demagnetized in 50 Oe steps from 50 to 200 Oe and that treatment producing the highest value of K (the Fisherian precision parameter, Fisher 1953), consistent with stability of the site magnetic directions was used to clean all specimens from the site. The optimum cleaning field was usually 150 Oe. Forty-eight specimens were heated in a field-free space to 625 °C from 100 °C in 75 °C steps. Structural corrections were made by rotating site magnetic vectors around the strike of the host rocks. This simple correction is valid as the basin rocks typically have homoclinal dips and show little folding around sub-vertical axes.

The following criteria were used for rejecting results:

(1) Clustering of directions about two or more means with increasing cleaning, usually associated with outcrops consisting of isolated boulders.

(2) Instability persisting up to 200 Oe.

(3) High NRM moments (2.5×10^{-2} emu/cm^2 or higher), which may be indicative of lightning strikes.

(4) Site α_{95} radii exceeding 10° after cleaning.

The results of 11 sites were rejected by the above criteria. Most specimens showed some streaking towards the present pole and most natural remanent magnetization (NRM) intensities ranged from 1×10^{-3} to 2.5×10^{-3} emu/cm^2. Site α_{95} after cleaning rarely exceeded 8°. Between-site dispersion due to glacial disruption and minor faulting remains in the data and cannot be corrected for, but large numbers of sites on each intrusion tend to statistically reduce the importance of these effects.

Connecticut Paleomagnetic Results

The 41 site VGP's of the 5 Connecticut dikes are listed in Table 2 and the mean poles of the dikes, compiled by averaging site VGP's, are listed in Table 3 and shown in Fig. 3. The pole positions of the Bridgeport and Higganum dikes, both outcropping in the piedmont, are only 4° apart and the angular divergence between them is not statistically significant. The pole positions of the other dikes are statistically indistinguishable from the Bridgeport and Higganum positions when no structural corrections for the other dikes are made. The value of K is increased significantly (Cox 1969) for the mean poles of the basin dikes when no structural corrections are made. VGP's of the Hartford Basin sites of the Buttress dike are closer to those of the piedmont part if no correction for the basin sites is made. Thus it is apparent that the narrow dikes were intruded after the Hartford Basin rocks were tilted. The Cross Rock dike pole is displaced northeast of those of the other dikes, but the angular distance between its pole position and the others is not significant. These results do not agree with those of deBoer (1967), who calculated a mean pole position of 148°E, 65°N from 5 sites on the Bridgeport, Higganum, and Fairhaven dikes. The difference between the pole of deBoer (1967) and those of the same dikes listed in Table 3 is significant. This discrepancy may be the result of excess AF cleaning by deBoer (1967), leading to obliteration of most of the original thermoremanent magnetization (TRM) (J. deBoer, personal communication 1975).

The mean pole of the 41 VGP's of the Connecticut dikes is located at 99.7°E, 69.3°N, $\alpha_{95} = 2.2°$ (Table 3, Fig. 4). This position is 8° of arc away from an 'Upper Triassic' mean pole position of 94°E, 64°N calculated by Beck (1972) from diabase, lava flows, and arkose of the Newark trend of the Appalachians. This compilation, however, included only two narrow-dike poles: one from the Cheshire dikes (Buttress and Cross Rock dikes) of Connecticut (deBoer 1968) and the other from the Shelburne dike of Nova Scotia (Larochelle and Wanless 1966). The Connecticut dike pole is only 5° of arc away from the mean pole of the Gettysburg Basin sills (102.5°E, 65.0°N; Beck 1972), suggesting that the two sets of intrusive rocks were emplaced closely together in time.

Maryland Paleomagnetic Results

The VGP's of the Gettysburg sill (combined with results of Beck (1972) on the same body), the Frederick dikes, and the Boyds sill are given in Table 2 and the mean poles of these intrusives are listed in Table 3 and shown in Fig. 3. The mean pole of the Frederick dikes is statistically indistinguishable from that of the Gettysburg sill when structural corrections for the Frederick sites, where appropriate, are made. Faill (1973)

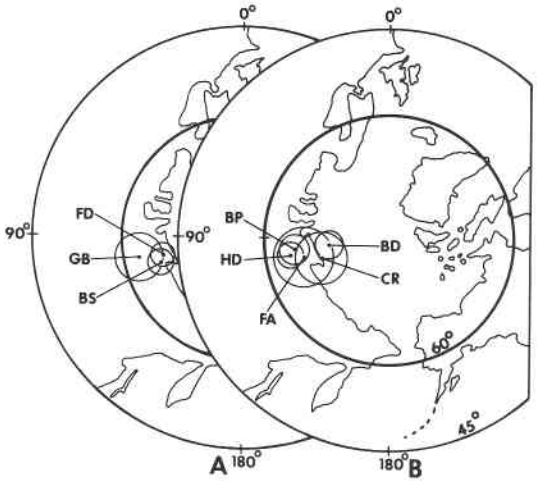


FIG. 3. A. Mean pole positions with α_{95} circles for the Gettysburg sill (including data from this body calculated by Beck 1972), Frederick dikes, and the Boyds sill of Maryland and southern Pennsylvania. The Gettysburg and Frederick pole positions are based on VGP's that have been corrected for structure. FD = Frederick dikes. GB = Gettysburg sill. BS = Boyds sill.

B. Mean poles with α_{95} circles for the Connecticut dikes. No structural corrections have been applied to any of the constituent VGP's of these poles. BP = Bridgeport dike. HD = Higganum dike. BD = Buttress dike. CR = Cross Rock dike. FA = Fairhaven dikes.

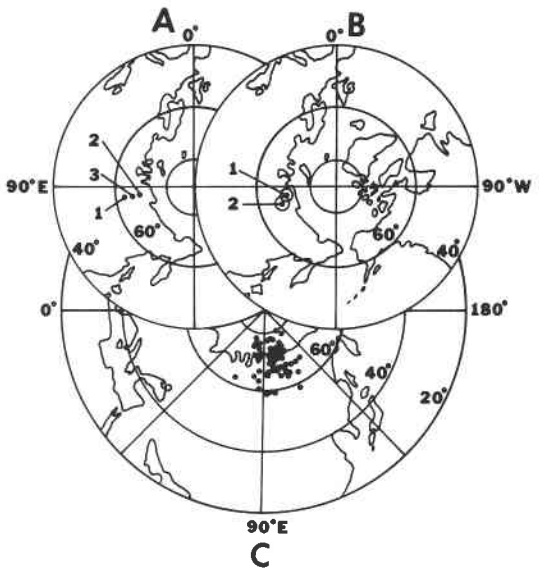


FIG. 4. A.(1) Mean pole of the Gettysburg Basin sills of Pennsylvania (Beck 1972). (2) Mean pole of the 68 VGP's of the Connecticut, Maryland, and southern Pennsylvania dikes and sills (including 7 VGP's from the Gettysburg sill calculated by Beck 1972). (3) Mean pole of the 139 VGP's calculated by Beck (1972) and this study. Alpha-$_{95}$ circles about these pole positions are very small and have been omitted for clarity.

B.(1) Mean pole position of the 42 reliable VGP's of Connecticut. (2) Mean pole of the 27 reliable VGP's of Maryland and southern Pennsylvania, including 7 calculated by Beck (1972) from the Gettysburg sill.

C. Distribution of the 61 reliable VGP's calculated in this work.

has shown that the Gettysburg sill was probably intruded before the basin rocks were tilted. Frederick sites in the piedmont, but close to the basin edge, were corrected for half the amount of dip of the basin rocks, as the east margin of the Gettysburg Basin is typically an onlap unconformity. The value of K for the Frederick mean pole is increased by nearly 100 when structural corrections for sites in the basin rocks and near the basin edge are made. Corrections were applied to all sites except FD-1, which is located 10 km east of the Culpeper Basin.

The paleomagnetic results of the Boyds sill are unusual in that its uncorrected mean pole position is much closer to those of the apparently comagmatic Gettysburg Basin sills than the structurally corrected pole. The corrected position is significantly removed from those of the Frederick dikes or the Gettysburg Basin sills. Aeromagnetic data (Blanchett *et al.* 1963) suggest that the Boyds sill is a stock. The similarity of the Boyds pole to those of the Gettysburg Basin sills, when no correction is applied to the Boyds pole, suggests that the Boyds sill was intruded after the Culpeper Basin rocks were tilted. For this reason, the uncorrected VGP's from the Boyds sill were used in computing a mean pole located at 103.3°E, 66.7°N, α_{95} = 2.4° (Table 3, Fig. 4) from diabase in Maryland and southern Pennsylvania. This pole position is only 3° from the mean pole of the Connecticut dikes, which is not significant at the 95% level.

Composite Pole Position

The mean pole of the 61 VGP's calculated in this paper is located at 100.9°E, 68.6°N, α_{95} = 1.6° (Table 3) and combined with 7 VGP's from the Gettysburg sill (Beck 1972), is located at 101.2°E, 68.3°N, α_{95} = 1.6° (Table 3, Fig. 4). These positions are in good agreement with pole positions calculated by Opdyke (1961), Larochelle and Wanless (1966), deBoer (1968), and Beck (1972) from diabase intrusions of the Newark trend (Table 1). Cornet *et al.* (1973) and Cornet and Traverse (1975) have shown

TABLE 2. Site paleomagnetic data from diabase intrusive rocks of Connecticut and Maryland

Site	Long. E	Lat. N	N	H	K	α_{95}	No structural correction				Structural correction				Dip direction	Dip
							Dec.	Inc.	Long.	Lat.	Dec.	Inc.	Long.	Lat.		
Buttress Dike																
BD-1	73°02′	41°17′	7	150	270	3.7	009.2	50.6	70.5	77.0	NA	NA	NA	NA	NA	NA
BD-2	72°59′	41°20′	8	200	470	2.8	000.3	36.0	106.2	67.9	NA	NA	NA	NA	NA	NA
BD-3	72°59′	41°20′	7	200	458	2.9	006.5	40.0	89.1	70.0	NA	NA	NA	NA	NA	NA
BD-4*	73°01′	41°17′	14	100	46	6.0	026.2	51.6	34.5	67.1	NA	NA	NA	NA	NA	NA
BD-5	72°55′	41°26′	9	150	71	6.1	007.2	47.2	82.3	75.2	036.8	48.6	28.7	57.9	115°	25°
BD-6	72°55′	41°26′	9	150	109	4.4	001.1	55.1	99.7	83.5	041.1	56.9	12.1	58.4	115°	25°
BD-7	72°52′	41°23′	9	150	85	5.6	002.7	50.3	95.7	78.8	021.5	56.2	38.3	72.7	120°	25°
BD-8	72°55′	41°24′	14	150	141	3.3	000.1	40.5	107.7	71.1	023.3	44.3	50.6	65.6	110°	25°
BD-9	72°53′	41°28′	15	200	850	1.3	009.7	51.8	72.2	78.5	039.2	54.5	17.7	58.8	120°	25°
BD-10	72°49′	41°31′	13	150	103	4.1	000.8	41.1	105.6	71.5	019.4	44.6	65.6	70.0	110°	20°
Higganum Dike																
HD-1	72°46′	41°19′	11	150	95	4.7	010.2	33.0	84.9	64.4	NA	NA	NA	NA	NA	NA
HD-2	72°44′	41°21′	7	150	39	9.8	006.3	35.9	92.4	67.3	NA	NA	NA	NA	NA	NA
HD-3	72°33′	41°28′	7	150	224	4.0	006.4	27.0	94.7	61.7	NA	NA	NA	NA	NA	NA
HD-4	72°33′	41°28′	8	150	668	2.3	009.7	26.5	88.5	60.8	NA	NA	NA	NA	NA	NA
HD-5	72°32′	41°29′	9	150	927	1.7	001.1	35.3	105.3	67.5	NA	NA	NA	NA	NA	NA
HD-6	72°32′	41°29′	11	150	299	2.6	359.7	30.9	108.7	64.7	NA	NA	NA	NA	NA	NA
HD-7	72°32′	41°29′	10	150	346	2.6	358.6	34.4	115.9	66.9	NA	NA	NA	NA	NA	NA
HD-8	72°32′	41°29′	9	150	167	4.0	353.8	34.5	122.7	66.4	NA	NA	NA	NA	NA	NA
HD-9	72°33′	41°29′	10	150	229	3.2	356.8	32.2	115.3	65.3	NA	NA	NA	NA	NA	NA
HD-10	72°27′	41°38′	21	150	62	4.1	001.5	33.0	103.7	70.3	NA	NA	NA	NA	NA	NA
HD-11*	72°28′	41°37′	12	150	11	13.6	314.9	4.5	165.4	33.7	NA	NA	NA	NA	NA	NA
HD-12	72°22′	41°43′	12	150	33	7.7	008.5	31.3	89.3	63.7	NA	NA	NA	NA	NA	NA
HD-13*	72°20′	41°44′	15	150	3	23.7	277.6	−13.0	−173.0	1.2	NA	NA	NA	NA	NA	NA
Bridgeport Dike																
BP-1	73°19′	41°11′	9	150	225	3.4	003.4	30.4	91.2	64.1	NA	NA	NA	NA	NA	NA
BP-2	73°15′	41°13′	9	150	388	2.6	002.6	28.1	94.2	62.3	NA	NA	NA	NA	NA	NA
BP-3	73°15′	41°12′	15	150	42	6.2	007.8	38.6	87.7	68.7	NA	NA	NA	NA	NA	NA
BP-4	73°15′	41°12′	6	150	70	9.2	010.0	41.3	80.3	69.8	NA	NA	NA	NA	NA	NA
BP-5*	73°15′	41°13′	9	150	3	36.0	323.1	12.9	161.2	41.8	NA	NA	NA	NA	NA	NA
BP-6	73°12′	41°15′	10	150	180	3.6	002.8	37.9	100.5	69.1	NA	NA	NA	NA	NA	NA
BP-7*	73°12′	41°25′	5	150	2	89.4	001.4	18.6	105.2	57.3	NA	NA	NA	NA	NA	NA
BP-8	73°06′	41°23′	7	150	341	3.3	006.8	43.2	87.2	75.3	NA	NA	NA	NA	NA	NA
BP-9*	73°10′	41°21′	8	150	6	26.2	081.5	70.0	−25.0	37.3	NA	NA	NA	NA	NA	NA
BP-10	72°55′	41°32′	5	150	280	4.6	358.2	29.3	111.9	63.6	NA	NA	NA	NA	NA	NA
BP-11	72°55′	41°32′	6	150	179	5.0	001.6	28.8	104.5	63.3	NA	NA	NA	NA	NA	NA

TABLE 2 (*Concluded*)

Site	Long. E	Lat. N	N	H	K	α_{95}	No structural correction				Structural correction				Dip direction	Dip
							Dec.	Inc.	Long.	Lat.	Dec.	Inc.	Long.	Lat.		
Fairhaven Dikes																
FA-1	72°53′	41°16′	11	100	85	5.0	358.4	34.7	110.8	67.5	013.3	37.9	74.3	66.9	105°	20°
FA-2	72°52′	41°18′	11	100	218	3.1	009.7	29.4	087.5	62.5	021.4	31.0	65.5	58.8	110°	20°
FA-3	72°51′	41°20′	9	200	241	3.3	021.4	33.7	63.8	60.3	033.9	36.9	44.3	54.3	115°	25°
FA-4*	72°49′	41°23′	6	150	9	21.6	328.5	76.9	−98.7	60.8	087.8	74.2	−34.2	36.6	115°	25°
FA-5	72°49′	41°23′	9	150	179	3.9	357.4	51.3	120.5	79.8	035.1	61.2	64.6	5.5	125°	25°
FA-6	72°49′	41°23′	9	150	162	4.0	356.9	43.7	117.6	73.6	026.4	55.6	69.6	26.4	125°	25°
FA-7	72°47′	41°25′	7	150	72	7.2	354.2	20.7	188.8	58.2	039.4	37.5	102.2	65.2	120°	25°
FA-8	72°47′	41°26′	10	150	438	2.3	002.7	32.2	100.7	65.4	020.5	40.3	57.8	64.4	120°	20°
Cross Rock Dike																
CR-1	72°56′	41°28′	8	100	347	3.0	004.8	38.7	95.1	67.4	019.1	41.1	61.3	65.7	110°	20°
CR-2	72°53′	41°27′	6	150	231	4.4	354.1	47.2	129.3	75.5	017.4	55.7	35.5	75.4	120°	20°
CR-3	72°52′	41°28′	6	150	245	4.3	353.7	49.9	134.3	77.6	027.6	58.7	15.6	69.0	120°	25°
CR-4	72°52′	41°28′	7	150	284	3.6	000.4	49.2	106.3	78.1	032.8	55.4	20.5	63.9	120°	25°
CR-5	72°51′	41°27′	10	250	49	6.9	003.9	30.3	98.1	64.5	020.2	36.0	62.4	62.5	115°	25°
CR-6	72°50′	41°27′	11	150	42	7.1	347.5	38.1	130.8	65.9	003.8	49.7	91.8	78.1	120°	20°
Boyds Sill																
BS-1	77°20′	39°12′	5	100	3890	1.2	352.0	27.6	119.7	63.7	348.5	21.2	124.6	59.2	300°	15°
BS-2	77°20′	39°12′	5	100	413	3.8	357.7	28.2	107.2	64.9	353.7	22.6	114.7	61.2	300°	15°
BS-3	77°20′	39°11′	5	100	668	3.0	359.1	31.3	104.2	66.9	354.5	25.8	113.9	63.2	300°	10°
BS-4*	77°21′	39°12′	8	150	66	6.8	295.4	6.6	178.0	21.6	295.4	−5.3	173.3	17.7	300°	10°
BS-5*	77°21′	39°11′	11	150	3	35.2	009.6	−16.9	89.3	41.5	013.5	−20.7	84.9	38.7	300°	10°
BS-6*	77°21′	39°11′	13	150	15	10.9	353.6	−15.2	110.7	42.9	356.9	−22.0	105.9	39.5	300°	10°
BS-7	77°20′	39°12′	8	100	217	3.8	357.5	42.4	110.4	74.4	349.7	35.2	128.2	67.6	300°	10°
BS-8	77°21′	39°12′	7	100	498	2.7	003.4	33.5	93.4	68.1	357.2	27.5	108.3	64.5	300°	10°
BS-9	77°20′	39°11′	8	50	292	3.2	359.2	30.6	103.8	66.5	354.0	23.9	114.5	62.0	300°	10°
BS-10	77°21′	39°11′	6	100	402	3.3	357.3	28.6	108.1	65.1	352.7	21.6	116.7	60.5	300°	10°
Frederick Dikes																
FD-1	77°23′	39°19′	12	200	77	5.3	356.8	43.8	113.6	71.8	NA	NA	NA	NA	NA	NA
FD-2	77°20′	39°27′	14	150	912	1.3	007.3	29.6	85.2	65.2	002.1	26.9	97.4	64.2	290°*	10°**
FD-3*	77°20′	39°28′	8	150	16	14.5	060.4	71.7	−32.0	48.1	029.9	75.9	−50.9	60.7	290°**	10°**
FD-4	77°18′	39°31′	12	150	56	5.8	003.1	42.5	91.4	74.4	355.1	38.9	116.6	71.5	290°**	10°**
FD-5	77°18′	39°31′	8	200	42	8.6	013.2	41.9	63.4	70.7	004.7	40.0	88.2	72.3	290°**	10°**
FD-6	77°18′	39°31′	16	200	67	4.5	002.5	40.5	93.4	72.9	355.2	36.9	115.3	70.2	290°**	10°**
FD-7	77°18′	39°35′	17	150	348	1.9	007.5	44.0	75.9	74.5	359.3	33.0	103.9	68.0	320°	15°
FD-8	77°18′	39°36′	20	150	229	2.2	008.2	37.7	79.4	69.6	359.7	28.3	102.8	65.0	310°	25°
FD-9	77°19′	39°37′	22	150	180	2.3	009.3	47.7	76.4	64.8	356.9	38.4	111.0	71.5	310°	15°
FD-10	77°40′	39°40′	14	150	474	1.8	009.9	41.5	71.6	71.8	359.7	32.7	102.7	67.8	315°	15°
FD-11	77°15′	39°42′	17	150	196	2.5	016.7	38.4	59.2	66.8	003.6	20.1	91.7	61.0	335°	25°
Gettysburg Sill																
GS-1	77°18′	39°48′	8	200	542	2.4	012.6	31.3	73.1	64.5	005.4	18.4	91.6	59.1	320°	20°
GS-2	77°15′	39°48′	7	200	699	2.3	008.6	33.0	80.8	66.9	001.6	18.9	98.9	59.7	320°	20°
GS-3	77°18′	39°49′	8	200	1116	1.7	010.0	32.7	77.8	66.1	002.6	19.4	96.9	59.9	320°	20°

NOTE: N = Number of specimens. H = Level of demagnetizing field in oersteds. K = Fisherian (1953) precision parameter. α_{95} = Radius of cone of 95% confidence. Dec = Declination of the site mean magnetic vector. Inc = Inclination of the site mean magnetic vector. Dip direction = Azimuth of dip. Dip = Dip of host rocks.
*Results from this site were rejected for reasons given in the text. NA = Not applicable.
**Estimated structural correction.

TABLE 3. Intrusion and regional mean poles

Intrusion	VGP's	No structural correction					Structural correction				
		K	α_{95}	S	Long.	Lat.	K	α_{95}	S	Long.	Lat.
Higganum	11	188	3.3	5.9	99.8	65.5	NA	NA	NA	NA	NA
Bridgeport	8	214	3.8	5.6	95.5	66.9	NA	NA	NA	NA	NA
Buttress	9	199	3.7	5.7	92.0	74.9	34	8.9	13.8	46.2	68.7
Fairhaven	7	51	8.5	11.3	100.1	68.0	33	10.7	14.0	54.0	66.1
Cross Rock	6	98	6.8	8.2	113.9	72.5	57	9.0	10.8	45.2	71.3
Connecticut mean pole	41	102	2.2	8.0	99.7	69.3	(No structural correction for any site VGP's)				
Boyds	7	314	3.4	4.6	107.4	67.3	370	3.0	4.2	116.3	62.9
Frederick	10	169	3.7	6.2	78.7	72.3	284	3.0	4.8	103.7	69.0
Gettysburg	3	1699	3.0	1.9	77.1	65.8	1703	3.0	1.9	95.8	59.6
Gettysburg*	10	52	6.1	11.3	79.4	66.8	72	5.7	9.5	101.5	63.2
*Maryland–southern Pennsylvania mean pole**	27	127	2.4	7.2	103.3	66.7	(Structural correction for site VGP's from the Frederick and Gettysburg bodies)				
Mean pole of this paper	61	113	1.6	7.6	100.9	68.6	NA	NA	NA	NA	NA
*Connecticut, Maryland, and southern Pennsylvania mean pole**	68	108	1.6	7.8	101.2	68.3	NA	NA	NA	NA	NA
Mean pole of Connecticut, Pennsylvania, and Maryland intrusions	139	112	1.3	7.7	101.6	65.4	NA	NA	NA	NA	NA

K = Fisherian (1953) precision parameter. α_{95} = Radius of cone of 95% confidence. S = Fisherian (1953) angular standard deviation parameter.
*Combined with results of Beck (1972) from the Gettysburg sill. NA = Not applicable.

that the upper rocks in the Hartford and Gettysburg Basins contain Early to early Middle Jurassic pollen assemblages. The Connecticut narrow dikes, the Gettysburg Basin sills, and the Frederick dikes intrude either the youngest rocks of the basins or structures involving the youngest rocks and may therefore be of Early Jurassic age. The mean pole given above and that of the Gettysburg Basin sills may be Early Jurassic paleopoles on the basis of the indicated pollen age. The apparent different time of basin downwarping relative to intrusion of the Gettysburg Basin sills, the Frederick dikes, and the Connecticut dikes supports the conclusions of Klein (1969) and Faill (1973) that the Newark Group basins developed as separate basins with different structural histories.

Combining the results of this study and those of Beck (1972) yields a mean pole position of 101.6°E, 65.4°N, $\alpha_{95} = 1.3°$ from 139 sites (Table 3). This pole position is shown in Fig. 4 and may be representative of the pole position for North America during Early Jurassic time. The large number of intrusive units contributing to this position probably ensures that the effects of secular variation are included in the dispersion of VGP's about the mean position, shown in Fig. 5. The distribution of VGP's is nearly circular, which may be the result of little or no apparent polar motion during the time span represented by the Connecticut, Pennsylvania, and Maryland intrusive rocks. Azimuthal and latitudinal symmetry tests using the X^2 statistic (Watson and Irving 1957) indicate that the distribution of VGP's is Fisherian at the 95% confidence level. The small scatter of VGP's evident in Fig. 5 is suggestive of low magnitude of PSV or smoothing of rock magnetic directions during cooling of the intrusions. The Fisherian angular standard deviation estimate

$$s = 81/\sqrt{K}$$

which is most commonly used to describe the magnitude of PSV (Brock 1971) is listed in Table 3 for each of the intrusive and regional mean poles. Baag and Helsley (1974) have shown that expected values for s from distributions of VGP's should range between 14 and 20° for latitudes between 0 and 25°, corresponding to the paleolatitudes of the Connecticut, Pennsylvania, and Maryland intrusions. All values of s, whether for intrusive or regional mean poles, are less than

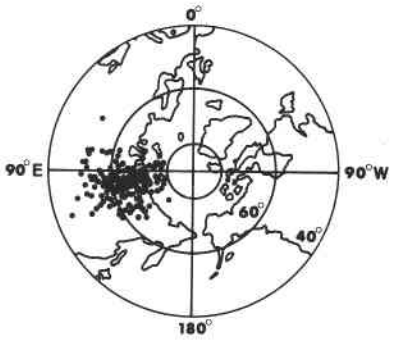


FIG. 5. Distribution of the 139 VGP's calculated by Beck (1972) and in this paper from dikes and sills in Connecticut, Pennsylvania, and Maryland.

14°. Beck (1972) suggested that low PSV is characteristic of the Late Triassic – Early Jurassic paleofield. However, Brock (1971, Fig. 4) has shown that PSV was probably of normal magnitude during this time period. Similarly, Dalrymple *et al.* (1975) have shown that the dispersion of VGP's about the mean pole of Lower Jurassic diabase dikes and sills of Liberia is of expected magnitude. These findings suggest that the low values of s are the result of magnetic smoothing processes.

The Curie temperatures of 48 specimens ranged between 475 and 625 °C. Opaque microscopy of 49 polished sections revealed generally low oxidation states, most in classes I and II of Wilson and Watkins (1967). Wilson *et al.* (1968) concluded that low oxidation states are indicative of only slight high-temperature (1100–500 °C) deuteric oxidation. Most of the sections showed faint patchy intergrowths of titanomaghemite and martite and broad ilmenite lamellae. Many of the specimens showed a slight peak in spontaneous magnetization on heating to 400 or 475 °C (Fig. 6); Ozima and Ozima (1971) and Johnson and Merrill (1973) have indicated that such behavior is typical of rocks containing titanomaghemite. Stacey and Bannerjee (1974) stated that titanomaghemite is formed by low temperature (about 150–300 °C) oxidation of titanomagnetite. Sanver and O'Reilly (1970), Carmichael and Nicholls (1967), Readman and O'Reilly (1971), and Johnson and Merrill (1973) have shown that fresh basalts with unaltered titanomagnetite have Curie temperatures of 200–250 °C, whereas similar rocks which have undergone low temperature oxidation in laboratory conditions have

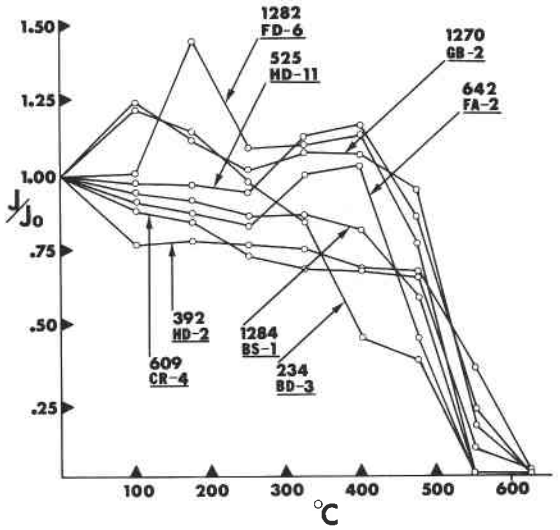


FIG. 6. Thermoremanent decay curves for 8 representative specimens. Specimen numbers and site localities are given for each curve.

Curie temperatures of 400–600 °C. Johnson and Merrill (1973) showed that the change in Curie temperatures is the result of thermal breakdown of titanomaghemite produced by low-temperature oxidation into magnetite and ilmenite. Stacey and Bannerjee (1974) stated that this breakdown occurs near 350–450 °C, which may account for the 'humps' in the thermal decay curves in this temperature region. The formation of titanomaghemite is important in that its spontaneous magnetization is similar to that of titanomagnetite, but the direction of its chemical remanent magnetization (CRM) is not necessarily faithful to the TRM directions of the host titanomagnetite (Johnson and Merrill 1973). The results of long-term growth of titanomaghemite could be the gradual erasure of dispersion due to secular variation.

Beck (1972) considered that CRM averaging effects would produce site VGP's with large dispersion; however, if averaging effects took place in volumes smaller than those of single specimens, sites with uniformly averaged directions would be characterized by small dispersion and large *K*. Cornet *et al.* (1973) and Cornet and Traverse (1975) have shown that the Newark Group basins underwent only moderate burial heating up to a maximum of about 150–200 °C, but probably high enough to cause oxidation of titanomagnetite to titanomaghemite (Stacey and Bannerjee 1974). Faill (1973) and Dickenson (1974) have shown that the Newark Group basins are probably relics of an early Atlantic rifting episode and therefore the basin–piedmont region probably had steep thermal gradients, such that rocks at depths of 2–6 km may have been subjected to temperatures of 100–200 °C, whether in the piedmont or in the basins. It is likely that this thermal regime lasted at least 10^5–10^6 years (Vine and Hess 1970), certainly greater than the period of secular variation (Baag and Helsley 1974); therefore a titanomaghemite CRM would have been able to average out the dispersion of directions due to secular variation.

Despite the apparent averaging of magnetic directions that occurred, it appears likely that the diabase pole positions are reliable in that they show no streaking towards Upper Jurassic poles given by Symons (1973) and Steiner (1974) or the present pole, and are characterized by circularly disposed VGP's. A process involving averaging of directions over 20–40 m.y. would perhaps produce an elliptical distribution of VGP's, caused by motion of the continent during the remagnetization of the rocks. It is therefore apparent that the averaging process lasted only about 1–10 m.y., which may correspond to the amount of time that elapsed between the conclusion of diabase intrusion and the cessation of the thermal regime associated with initial rifting along the Newark basin trend.

Reconstruction of Europe and North America

Phillips and Forsyth (1972) calculated a mean Jurassic pole position for North America of 174°E, 64°N. They concluded that this position was highly anomalous, as the rotation required to place North America against Europe in the pre-rifting arrangement of Bullard *et al.* (1965) increased the divergence between this pole position and that from the European Jurassic. This position, rotated according to the pre-rifting fit of Bullard *et al.* (1965), is located at 107°E, 64°N, only 6° of arc away from the mean pole of the Connecticut, Pennsylvania, and Maryland diabases that was calculated in this study (Fig. 7). This suggests that the diabase pole may be typical of at least the Lower Jurassic rocks for North America and that an alternate fit model proposed by Phillips and Forsyth (1972) is not necessary. Obtaining a mean pole position from the five reliable European Jurassic pole positions

TABLE 4. European Jurassic paleomagnetic pole positions used in calculating a European Jurassic mean pole

Location	Sites/Specimens	Long. E	Lat. N	α_{95}	Source
France					
Pyrenees volcanics	7/26	114.2	62.5	7.3	Listing of "reliable" pole positions in Hicken *et al.* (1972)
Pyrenees volcanics	8/26	143.6	64.9	7.0	
Austria					
Alpine red limestone	?/16	109.0	51.0	7.0	
Alpine radiolarite	?/15	122.0	56.0	6.0	Hicken *et al.* (1972)
Czechoslovakia					
mineralized veins	8/113	187.0	78.7	5.1	Hicken *et al.* (1972)
Mean European Jurassic pole	(5 VGP's)	130.9	64.3	14.6	

Long. = longitude of paleomagnetic pole. Lat. = latitude of paleomagnetic pole. α_{95} = radius of cone of 95% confidence.

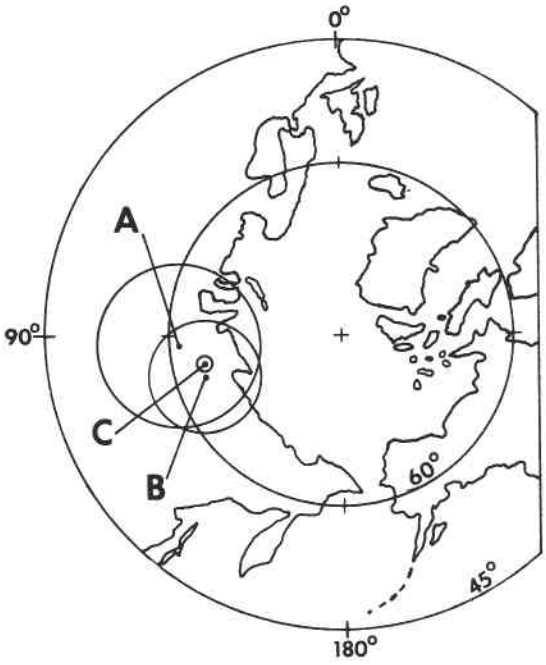


FIG. 7. Comparison of rotated European Jurassic mean poles and the mean pole of the Connecticut, Pennsylvania, and Maryland dikes and sills. A. European Jurassic mean pole calculated from reliable results listed in Hicken *et al.* (1972). B. European Jurassic mean pole calculated by Phillips and Forsyth (1972). C. Mean pole of the Connecticut, Pennsylvania, and Maryland dikes and sills. The European poles have been rotated according to the pre-rifting arrangement of Bullard *et al.* (1965).

listed in Hicken *et al.* (1972) (Table 4) and rotating it to close the Atlantic produces a mean European Jurassic pole position of 93°E, 64°N (Table 5). This position is somewhat removed from the Appalachian diabase pole position, but the distance between this European position and the North American Jurassic pole of Phillips and Forsyth (1972) is much greater. However, this European Jurassic pole is probably best considered an approximation owing to the small number of contributing poles (5) and the large dispersion between these poles. The angular divergence of both rotated European Jurassic poles and the Appalachian diabase pole based on the results of Beck (1972) and this paper is not significant (Fig. 7).

From the above information it appears that the Appalachian diabase pole based on the results of Beck (1972) and this paper is representative for the Lower Jurassic as well as the Upper Triassic rocks for North America and that the pre-rifting arrangement of Bullard *et al.* (1965) is probably accurate for North America and Europe. The coincidence of the European Jurassic and the Appalachian diabase poles when the continents are rotated together indicates that initial separation of Europe from North America had not begun by at least Early Jurassic time. This is in agreement with conclusions of Pitman and Talwani (1972), based on sea-floor magnetic anomalies, marine faunal assemblages, and paleomagnetism, that although the portion of the North Atlantic basin between Africa and North America (between the Guinea–Barracuda fracture zones and the Azores) began separating in Early to Middle Jurassic time, the portion of the basin between Europe and North America (north of the Azores) underwent only minor separation until Middle to Late Cretaceous time.

Conclusions

The following conclusions can be drawn from this study:

TABLE 5. Rotated European Jurassic mean poles

Unrotated pole			Rotation pole*			Rotated pole	
Long.	Lat.	α_{95}	Long.	Lat.	Rotation angle	Long.	Lat.
147.0	65.0**	11.0	27.7	88.5	38°	107.0	64.0
130.9	64.3†	14.6	27.7	88.5	38°	93.1	64.0

*The rotation pole and angle are from the Atlantic-closing model of Bullard *et al.* (1965).
**European Jurassic mean pole given by Phillips and Forsyth (1972).
†European Jurassic mean pole calculated from reliable European Jurassic paleomagnetic pole positions listed in Hicken *et al.* (1972).

(1) There is remarkably little difference among pole positions calculated from diabase dikes and sills which intrude the youngest rocks, or structures involving the youngest rocks, of the Gettysburg and Hartford Basins of the north-central Appalachian piedmont. These dikes and sills are probably of Early to early Middle Jurassic age and yield a mean pole position of 101.6°E, 65.4°N, $\alpha_{95} = 1.3°$ from 139 VGP's.

(2) The similar pole positions of the Connecticut dikes (with no structural corrections) and the Gettysburg Basin sills and dikes (corrected for structure) show that these basins did not subside simultaneously. This is in agreement with Faill (1973) and Klein (1969), who concluded that the Newark Group basins of the north-central Appalachians developed as isolated basins and not as a single once-contiguous basin (Sanders 1963).

(3) The small dispersion of VGP's around the mean pole of all VGP's is probably the result of magnetic 'homogenization' by low-temperature oxidation processes that may have persisted over a sufficiently long period of time (10^5–10^6 years) to average out the dispersion due to secular variation.

(4) The convergence of European Jurassic mean poles with the mean pole of the Connecticut dikes and the dikes and sills of Maryland and Pennsylvania when Europe is rotated against North America shows that significant separation of the two continents had not begun by Early to early Middle Jurassic time. The large α_{95} circles about the European Jurassic poles are probably the result of poor age control on the rocks that were sampled and the small amount of contributing paleomagnetic data.

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Anonymous. 1966. Aeromagnetic map of the Mount Carmel Quadrangle, Connecticut. U.S. Geol. Surv. Map GP-864.

BAAG, CZANG-GO and HELSLEY, C. E. 1974. Geomagnetic secular variation model E. J. Geophys. Res. **79**, pp. 4918–4922.

BECK, M. E. 1972. Paleomagnetism of Upper Triassic diabase from southeastern Pennsylvania: Further results. J. Geophys. Res. **77**, pp. 5673–5687.

BLANCHETT, J., TYSON, N. S., and MCGOWAN, E. F. 1963. Aeromagnetic map of the Germantown and part of the Poolesville Quadrangles, Montgomery and Frederick Counties, Maryland. U.S. Geol. Surv. Map GP-394.

BOWKER, D. E. 1960. Remanent magnetization of eastern United States Triassic rocks. Unpubl. PhD Diss., Mass. Inst. Technol., Cambridge, Massachusetts, 186 p.

BROCK, A. 1971. An experimental study of paleosecular variation. Geophys. J. R. Astron. Soc. **24**, pp. 303–317.

BULLARD, E., EVERETT, J. E., and SMITH, A. G. 1965. The fit of the continents around the Atlantic. R. Soc. London Phil. Trans., Ser. A258, pp. 41–51.

CARMICHAEL, I. and NICHOLLS, J. 1967. The iron–titanium oxides and oxygen fugacities in volcanic rocks. J. Geophys. Res. **72**, pp. 4663–4687.

CLOOS, E., FISHER, G. W., and HOPSON, C. A. 1964. The geology of Howard and Montgomery Counties, Maryland. State Board of Natural Resources, Baltimore, Maryland, 373 p.

CORNET, B. and TRAVERSE, A. 1975. Palynological contributions to the chronology and stratigraphy of the Hartford Basin in Connecticut and Massachusetts. Geosci. and Man, **11**, pp. 1–33.

CORNET, B., TRAVERSE, A., and MCDONALD, N. C. 1973. Fossil spores, pollen, and fish from Connecticut indicate Early Jurassic age for part of the Newark Group. Science, **182**, pp. 1243–1246.

COX, A. 1969. Confidence limits for the precision parameter K. Geophys. J. R. Astron. Soc. **18**. pp. 545–549.

DALRYMPLE, G. B., GROMME, C. S., and WHITE, R. W. 1975. Potassium–argon age of diabase dikes in Liberia: Initiation of central Atlantic rifting. Bull. Geol. Soc. Am. **86**, pp. 399–411.

DEBOER, J. 1967. Paleomagnetic–tectonic study of Meso-

zoic dike swarms in the Appalachians. J. Geophys. Res. **72**, pp. 2237–2250.

——— 1968. Paleomagnetic differentiation and correlation of the Late Triassic volcanic rocks in the Central Appalachians (with special reference to the Connecticut Valley). Bull. Geol. Soc. Am. **79**, pp. 609–626.

DICKENSON, W. R. 1974. Plate tectonics and sedimentary basins. San Joaquin Geol. Soc. short course: Geologic interpretation from global tectonics with application for California geology and petroleum exploration, February 1974, pp. 9.1–9.6.

FAILL, R. T. 1973. Tectonic development of the Triassic Newark–Gettysburg Basin in Pennsylvania. Bull. Geol. Soc. Am. **84**, pp. 725–740.

FISHER, SIR RONALD. 1953. Dispersion on a sphere. Proc. R. Soc. London, A 217, pp. 295–305.

FRITTS, C. E. 1963. Bedrock geology of the Mount Carmel Quadrangle. U.S. Geol. Surv. Quadrangle Map 199.

HICKEN, A., IRVING, E., LAW, L. K., and HASTIE, J. 1972. Catalogue of paleomagnetic directions and poles. Canada Dep. Energy, Mines, Resour., Ottawa, Canada, 135 p.

JOHNSON, H. P. and MERRILL, R. T. 1973. Low temperature oxidation of a titanomagnetite and the implications for paleomagnetism. J. Geophys. Res. **78**, pp. 4938–4949.

KLEIN, G. DE. 1969. Deposition of Triassic sedimentary rocks in separate basins, eastern North America. Bull. Geol. Soc. Am. **80**, pp. 1825–1832.

LAROCHELLE, A. and WANLESS, R. K. 1966. The paleomagnetism of a Triassic diabase dike in Nova Scotia. J. Geophys. Res. **71**, pp. 4040–4044.

MCELHINNEY, M. W. 1973. Paleomagnetism and plate tectonics. Cambridge Univ. Press, London, England. 358 p.

OPDYKE, N. D. 1961. The paleomagnetism of the New Jersey Triassic: A field study of the inclination error in red sediments. J. Geophys. Res. **66**, pp. 1941–1949.

OZIMA, M. and OZIMA, M. 1971. Characteristic thermomagnetic curves in oceanic basalts. J. Geophys. Res. **76**, pp. 2051–2056.

PHILLIPS, J. D. and FORSYTH, D. 1972. Plate tectonics, paleomagnetism, and the opening of the Atlantic. Bull. Geol. Soc. Am. **83**, pp. 1579–1600.

PITMAN, W. C., III and TALWANI, M. 1972. Sea-floor spreading in the North Atlantic. Bull. Geol. Soc. Am. **83**, pp. 1579–1600.

READMAN, P. W. and O'REILLY, W. 1971. Oxidation processes in titanomagnetite. Zeit. Geophys. **37**, pp. 329–338.

SANDERS, J. E. 1963. Late Triassic tectonic history of Northeastern United States. Am. J. Sci. **261**, pp. 501–524.

SANVER, M. and O'REILLY, W. 1970. Identification of naturally occurring non-stoichiometric titanomagnetites. Phys. Earth Planet. Int. **2**, pp. 166–174.

SMITH, R. C., ROSE, A. W., and LANNING, R. M. 1975. Geology and geochemistry of Triassic diabase in Pennsylvania. Bull. Geol. Soc. Am. **86**, pp. 945–955.

STACEY, F. D. and BANNERJEE, S. K. 1974. The physical principles of rock magnetism. Elsevier Sci. Publ. Co., Amsterdam, Netherlands, 195 p.

STEINER, M. B. 1974. Mesozoic apparent polar wander and Atlantic plate tectonics. Nature, **254**, pp. 107–109.

SYMONS, D. T. A. 1973. Paleomagnetic results from the Jurassic Topley Intrusions near Endako, British Columbia. Can. J. Earth Sci. **10**, pp. 1099–1108.

VINE, F. J., and HESS, H. H. 1970. Sea-floor spreading. *In*: The Sea. (A. Maxwell, *Ed.*) Vol. 4, Part III. John Wiley and Sons, New York, pp. 587–622.

WATSON, G. S. and IRVING, E. 1957. Statistical methods in rock magnetism. Mon. Not. R. Astron. Soc., Geophys. Supp. **7**, pp. 289–300.

WILSON, R. L., HAGGERTY, S. E., and WATKINS, N. D. 1968. Variation of paleomagnetic stability and other parameters in a vertical traverse of a single Icelandic lava. Geophys. J. R. Astron. Soc. **16**, pp. 79–96.

WILSON, R. L. and WATKINS, N. D. 1967. Correlation of petrology and natural magnetic polarity in Columbia Plateau basalts. Geophys. J. R. Astron. Soc. **12**, pp. 405–424.